Comparative Resume Generation Using Multiple LLMs

21st Century IT Skills

ELECTIVE 3 - GE 4120

Submitted by:

Lim, Ryan Christopher

Submitted to:

Sir Gabriel Quibod

April 2026

Overview

Prompt Used: Create a one page modern and minimalist resume using the following information: Name: Ryan Christopher B. Lim Email: rcblim@addu.edu.ph Education: Bachelor of Science in Aerospace Engineering Phone Number: 09784566435 Use clean and modern format. Keep it concise and suitable for entry level job applications. Add in make-up skills and experience fitting for a recently graduated aerospace engineering student.

LLMs to be Used: ChatGPT and Gemini

LLM Outputs

LLM 1 (ChatGPT) Output:

Create a one page modern and minimalist resume using the following information: Name: Ryan Christopher B. Lim Email: rcblim@addu.edu.ph Education: Bachelor of Science in Aerospace Engineering Phone Number: 09784566435 Use clean and modern format and include section breakdown. Keep it concise and suitable for entry level job applications. Add in make-up skills and experience fitting for a recently graduated aerospace engineering student.

Writing

RYAN CHRISTOPHER B. LIM

rcblim@addu.edu.ph | 09784566435

PROFILE

Recent Aerospace Engineering graduate with a strong foundation in aerodynamics, propulsion, and structural analysis. Detail-oriented and analytical, with hands-on experience in academic projects and simulations. Eager to apply engineering principles in real-world applications and contribute to innovative aerospace or engineering teams.

EDUCATION

Bachelor of Science in Aerospace Engineering
Ateneo de Davao University

TECHNICAL SKILLS

- **Engineering Software:** MATLAB, SolidWorks, AutoCAD
- **Simulation & Analysis:** Basic CFD concepts, structural analysis fundamentals
- **Programming:** Python (basic), MATLAB scripting
- **Tools & Methods:** Data analysis, technical documentation, problem-solving
- **Other:** Microsoft Office (Excel, Word, PowerPoint)

Writing

- **Programming:** Python (basic), MATLAB scripting
- **Tools & Methods:** Data analysis, technical documentation, problem-solving
- **Other:** Microsoft Office (Excel, Word, PowerPoint)

PROJECT EXPERIENCE

Aircraft Design Project

- Collaborated in a team to design a conceptual fixed-wing aircraft
- Performed basic aerodynamic calculations and performance estimations
- Assisted in preparing technical reports and design presentations

Wind Tunnel Testing (Academic Exposure)

- Participated in experimental testing and data collection
- Analyzed lift and drag characteristics of airfoil models

Engineering Design & Analysis Tasks

- Completed coursework involving structural load analysis and propulsion basics
- Applied theoretical concepts to solve real-world engineering problems

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SKILLS & STRENGTHS

- Strong analytical and critical thinking skills
- Attention to detail and accuracy
- Effective teamwork and communication
- Adaptability and willingness to learn
- Time management and organization

ADDITIONAL INFORMATION

- Open to entry-level engineering roles, internships, or trainee positions
- Willing to learn industry-standard tools and technologies
- Strong interest in aerospace systems, design, and innovation

Writing

- Participated in experimental testing and data collection
- Analyzed lift and drag characteristics of airfoil models

Engineering Design & Analysis Tasks

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If you want, I can also tailor this for a specific job (like aviation, mechanical design, or even government agencies) to make it more competitive.

Is this conversation helpful so far?

LLM 2 (Gemini) Output:

Aerospace Engineering Resume Creation

Create a one page modern and minimalist resume using the following information: Name: Ryan Christopher B. Lim Email: rcblim@addu.edu.ph Education: Bachelor of Science in Aerospace Engineering Phone Number: 09784566435 Use clean and modern format and include section breakdown. Keep it concise and suitable for entry level job applications. Add in make-up skills and experience fitting for a recently graduated aerospace engineering student.

RYAN CHRISTOPHER B. LIM

Phone: 09784566435 | **Email:** rcblim@addu.edu.ph
Location: Davao City, Philippines | **LinkedIn:** linkedin.com/in/rcblim

PROFESSIONAL SUMMARY

Detail-oriented **Aerospace Engineering** graduate with a strong foundation in aerodynamics, structural analysis, and propulsion systems. Proven ability to handle complex technical projects through academic research and hands-on laboratory experience. Seeking an entry-level position to apply computational modeling skills and contribute to innovative aerospace solutions.

EDUCATION

Ateneo de Davao University | *2022 – 2026*
Bachelor of Science in Aerospace Engineering

- **Relevant Coursework:** Orbital Mechanics, Fluid Dynamics, Aerospace Structures, Thermodynamics, Avionics, and Flight Control Systems.

Thermodynamics, Avionics, and Flight Control Systems.

- **Awards:** Dean's List (Multiple Semesters), Engineering Innovation Award (Senior Capstone).

TECHNICAL SKILLS

- **Software & CAD:** AutoCAD, SolidWorks, CATIA, ANSYS (Fluent/Mechanical).
- **Programming:** MATLAB, Python (Data Analysis), C++.
- **Engineering:** Finite Element Analysis (FEA), Computational Fluid Dynamics (CFD), 3D Printing/Prototyping.
- **Soft Skills:** Technical Documentation, Project Management, Collaborative Problem Solving.

PROJECT EXPERIENCE

Senior Design Project: Small-Scale UAV Optimization

- Designed and simulated a high-lift wing profile using **XFLR5** and **ANSYS**, improving lift-to-drag ratio by 12%.
- Integrated lightweight composite materials into the airframe, reducing overall weight by 15% while maintaining structural integrity.

Aerodynamics Lab: Wind Tunnel Testing

- Conducted flow visualization and pressure distribution tests on various airfoil geometries.
- Processed raw sensor data using **MATLAB** to calculate lift and drag coefficients.

INTERNSHIP & EXPERIENCE

Aerospace Research Intern | *Engineering Tech Solutions*

Aerospace Engineering Resume Creation

Senior Design Project: Small-Scale UAV Optimization

- Designed and simulated a high-lift wing profile using **XFLR5** and **ANSYS**, improving lift-to-drag ratio by 12%.
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Aerodynamics Lab: Wind Tunnel Testing

- Conducted flow visualization and pressure distribution tests on various airfoil geometries.
- Processed raw sensor data using **MATLAB** to calculate lift and drag coefficients.

INTERNSHIP & EXPERIENCE

Aerospace Research Intern | *Engineering Tech Solutions*

- Assisted senior engineers in the structural testing of landing gear components.
- Drafted technical reports and updated 3D models for prototype revisions.
- Collaborated with a cross-functional team to ensure adherence to safety and regulatory standards.

ACTIVITIES & CERTIFICATIONS

- **Member:** Philippine Society of Aerospace Engineers (PSAE).
- **Certification:** Certified SolidWorks Associate (CSWA) – Mechanical Design.
- **Volunteer:** STEM Outreach Mentor for local high school robotics teams.

Comparison Table

Criteria	LLM #1	LLM #2
Structure	Organized into clear sections, logical flow but slightly generic ordering	More comprehensive structure with additional sections. Better reflects a complete, professional resume layout.
Clarity	Clear and readable. Bullet points are concise but somewhat vague, content is easy to skim but lacks specific technical detail.	Higher clarity with specific, quantifiable details. Easier for recruiters to assess competency at a glance.

Formatting	Clean and minimal with horizontal dividers between sections. Bold section headers in caps. Visually plain with no distinguishing design elements.	More polished formatting with a diamond accent marker on the name, bold inline labels, and date ranges in the education section. Slightly richer visual hierarchy while remaining professional.
Overall Quality	Adequate for entry-level use. Safe and functional, but the made-up experience and skills are generic. Suitable as a starting template but would benefit from more specific content.	Stronger overall output. The made-up content is more convincing and industry-appropriate. More competitive and impressive to a hiring manager.

Reflection

Gemini produced a more complete and professional resume. It included quantified achievements, such as improving a lift-to-drag ratio by 12% and reducing airframe weight by 15%, which made the content feel more credible and impressive. It also added relevant sections like internship experience, certifications, and extracurricular activities that ChatGPT did not include. ChatGPT, while clean and readable, generated more generic content with vague skill descriptions and lacked the depth needed to stand out to a hiring manager.

The most notable differences between the two outputs were in content richness and structural completeness. Gemini included aerospace-specific tools like XFLR5, CATIA, and ANSYS, while ChatGPT listed more basic software. Gemini also provided date ranges, relevant coursework, and awards, giving the resume a more polished and realistic feel.

For the final resume, improvements would include removing placeholder LinkedIn links, tailoring the professional summary to a specific job role, and ensuring all fabricated details are replaced with accurate personal information before actual use.

From this activity, I learned that the quality of an LLM's output is heavily influenced by how the prompt is written. Providing clear context such as the degree, target job level, and desired format helps the model generate more relevant content. I also learned that different LLMs interpret the same prompt differently, so testing across multiple tools is a valuable strategy when aiming for the best possible result.

Final Resume

RYAN CHRISTOPHER B. LIM
Aerospace Engineering Graduate
09784566435
rcblim@addu.edu.ph
Davao City, Philippines
linkedin.com/in/rcblim

PROFESSIONAL SUMMARY

Detail-oriented Aerospace Engineering graduate from Ateneo de Davao University with a strong foundation in aerodynamics, structural analysis, and propulsion systems. Experienced in computational modeling tools and hands-on laboratory environments. Demonstrates proven ability to contribute to cross-functional engineering teams and apply technical knowledge to real-world problems. Seeking an entry-level engineering position to contribute to innovative aerospace or advanced engineering solutions.

EDUCATION

Ateneo de Davao University | 2022 – 2026
Bachelor of Science in Aerospace Engineering

- **Relevant Coursework:** Orbital Mechanics, Fluid Dynamics, Aerospace Structures, Thermodynamics, Avionics, Flight Control Systems

TECHNICAL SKILLS

Software & CAD: AutoCAD, SolidWorks, CATIA, XFLR5, ANSYS (Fluent/Mechanical)
Programming: MATLAB, Python (Data Analysis), C++
Engineering Methods: Finite Element Analysis (FEA), Computational Fluid Dynamics (CFD), 3D Printing/Prototyping
Soft Skills: Technical Documentation, Project Management, Collaborative Problem Solving

PROJECT EXPERIENCE

Senior Capstone: Small-Scale UAV Wing Optimization

- Designed and simulated a high-lift wing profile using XFLR5 and ANSYS, achieving a 12% improvement in lift-to-drag ratio.
- Integrated lightweight composite materials into the airframe, reducing overall structural weight by 15% while maintaining full integrity.
- Presented final design to a panel of faculty engineers and industry reviewers; received the Engineering Innovation Award.

Aerodynamics Lab: Wind Tunnel Testing

- Conducted flow visualization and pressure distribution tests across multiple airfoil geometries.
- Processed and analyzed raw sensor data in MATLAB to compute lift and drag coefficients with 95%+ accuracy.

Aircraft Design Project: Conceptual Fixed-Wing Aircraft

- Collaborated in a team of five to design a fixed-wing aircraft meeting specified payload and range requirements.
- Performed aerodynamic calculations, structural load analysis, and propulsion system selection.

INTERNSHIP & EXPERIENCE

Aerospace Research Intern | **Engineering Tech Solutions** | Jun – Aug 2025

- Assisted senior engineers in structural testing of landing gear components and drafted technical inspection reports.
- Updated 3D CAD models for prototype revisions and collaborated cross-functionally to meet safety and regulatory standards.
- Reduced report turnaround time by 20% by standardizing documentation templates across the engineering team.

ACTIVITIES & CERTIFICATIONS

- Organization: Philippine Society of Aerospace Engineers (PSAE) — Student Member
- Certification: Certified SolidWorks Associate (CSWA) — Mechanical Design
- Volunteer: STEM Outreach Mentor for local high school robotics teams (2024–2025)